Project FOUR: Power exhaust modelling

In this project, an Alcator C-Mod edge plasma modelling case is provided. Deuterium is considered. The total power entering the Scrape-Off Layer is 3.6 MW. Please answer the following question according to the corresponding plots from the modelling results.

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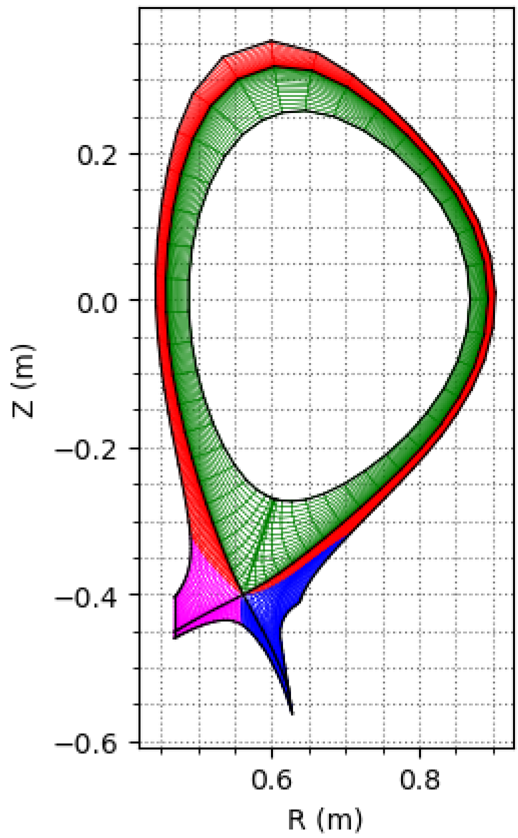
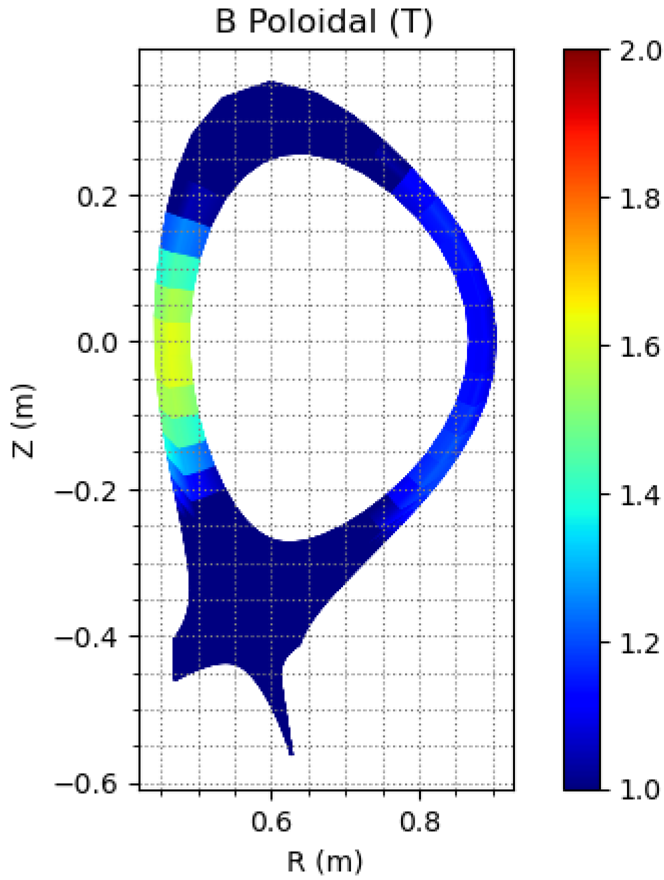
In [8]:

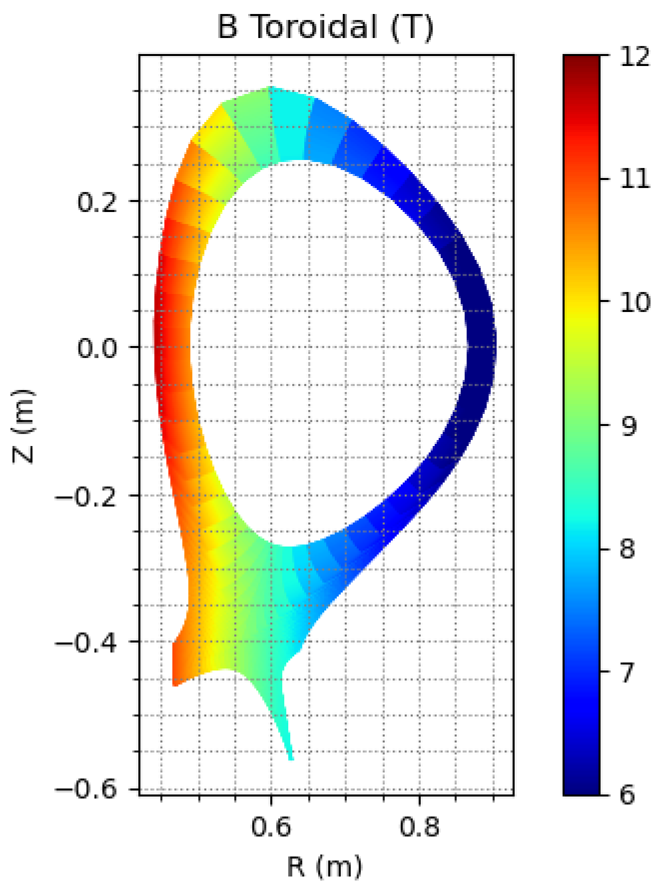
```
# In this part, we import necessary enviromental variables and functions,
# which are used in this hand-on session.
import matplotlib.pyplot as plt
import numpy as np
import math
import sys
sys.path.append('/home/jovyan/pex2024')
import sptools.read as spr
import sptools.plot as spp
```

In [9]:

```
CMod_b2fgmtry_file = './Data/b2fgmtry'
gmtry = spr.read_b2file(CMod_b2fgmtry_file)
spp.plot_geometry(gmtry)
B_pol = gmtry['bb'][:, :, 0].reshape(98, 38)
spp.plot_2dfield(gmtry,B_pol,figname='B Poloidal (T)',scale='linear')
B_tor = np.abs(gmtry['bb'][:, :, 2].reshape(98, 38))
spp.plot_2dfield(gmtry,B_tor,figname='B Toroidal (T)',scale='linear')
```


Z (m)
0.2
0.0
−0.2
−0.4
−0.6
0.6
0.8
R (m)


B Poloidal (T)
Z (m)
0.2
0.0
−0.2
−0.4
−0.6
0.6
0.8
R (m)
2.0
1.8
1.6
1.4
1.2
1.0



Question 1: The above plots are computational mesh and poloidal and toroidal magnetic field of Alcator C-Mod. 1) Please estimate the major and minor radius. 2) Please estimate the connection length and explain your reasons. 3) Please estimate the saft factor q and explain your reasons.

Answer

1) Estimation of Geometric Parameters

Based on the computational mesh and the magnetic field plots (Alcator C-Mod geometry), here is the refined estimation of the plasma parameters.

1. Major and Minor Radius

To find the physical dimensions of the plasma, I analyzed the cross-section at the midplane ($Z = 0$) using the mesh plot.

- **Inner Boundary:** The plasma starts on the high-field side at $R_{in} \approx 0.45\, m$.
- **Outer Boundary:** The plasma ends on the low-field side at $R_{out} \approx 0.90\, m$.

From these two points, I calculated the geometric center (R_0) and the plasma half-width (a):

$$R_0 = \frac{R_{out} + R_{in}}{2} = \frac{0.90 + 0.45}{2} = \mathbf{0.675\, m}$$

$$a = \frac{R_{out} - R_{in}}{2} = \frac{0.90 - 0.45}{2} = \mathbf{0.225\, m}$$

2. Connection Length (L_c)

The connection length is the distance a particle travels along a field line from the Outer Midplane (OMP) to the target.

Method: I estimated L_c by integrating the local field pitch. The formula relates the total path length to the poloidal projection:

$$L_c \approx \frac{B_{tor}}{B_{pol}} \times L_{pol}$$

Observations from plots:

- **Field Ratio:** At the OMP ($R \approx 0.9$ m), $B_{tor} \approx 6.0\,T$ and $B_{pol} \approx 1.2\,T$, giving a pitch ratio of ≈ 5.0.
- **Poloidal Length (L_{pol}):** I traced the separatrix path from the midplane ($Z = 0$) down to the divertor region. Identifying the **X-point** (intersection of the black axes) at approximately $R \approx 0.56\,m$ **and** $Z \approx -0.4\,m$, the curved path from the midplane to this point is roughly $0.6\,m$. Adding the short leg from the X-point to the target plate, the total poloidal distance is approximately $L_{pol} \approx 0.7\,m$.

Calculation:

$$L_c \approx 5.0 \times 0.7\,m \approx \mathbf{3.5\,m}$$

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3. Safety Factor (q)

The safety factor q represents the number of toroidal turns per poloidal turn. Since the plasma is vertically elongated, I used the formula that accounts for elongation (κ):

$$q \approx \frac{aB_{tor}}{R_0 B_{pol}} \left(\frac{1+\kappa^2}{2} \right)$$

Estimation of parameters:

- **Aspect Ratio term (a/R_0):** $0.225/0.675 \approx 0.33$.
- **Field Ratio (B_{tor}/B_{pol}):** ≈ 5.0.
- **Elongation (κ):** Using the X-point coordinate ($Z \approx -0.4\,m$) as a reference for the vertical extent, and assuming symmetry, the plasma height b is approximately $0.4\,m$.

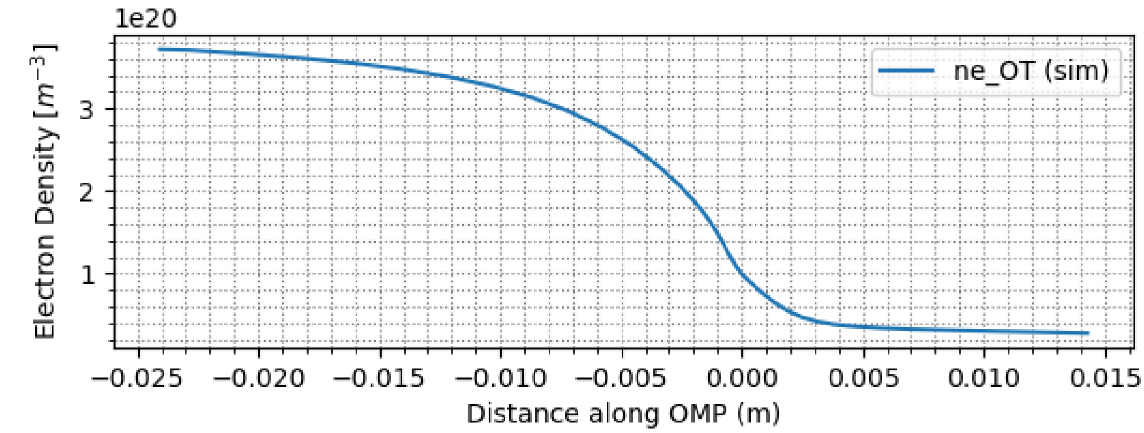
$$\kappa = \frac{b}{a} \approx \frac{0.4}{0.225} \approx 1.78$$

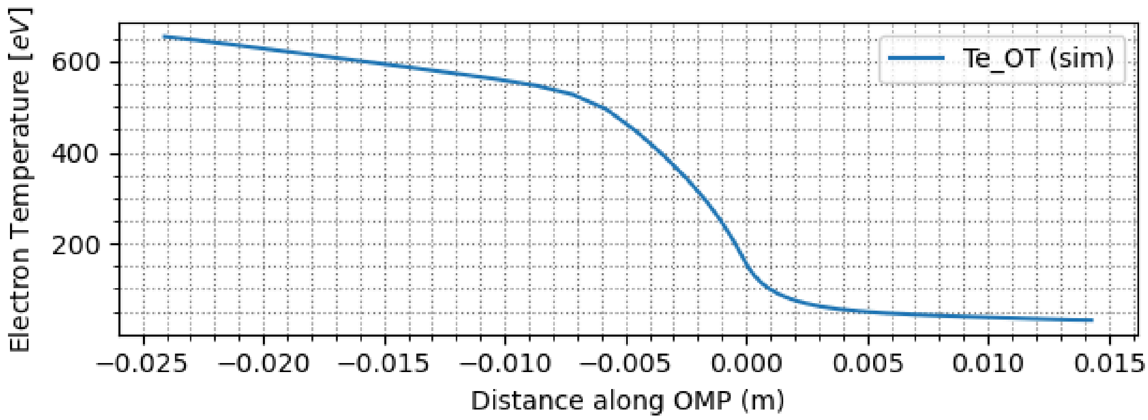
Calculation:

$$q \approx (0.33) \times (5.0) \times \left(\frac{1+1.78^2}{2} \right) \approx 1.65 \times 2.08 \approx \mathbf{3.4}$$

Conclusion: A safety factor of $q \approx 3.4$ indicates a robust configuration for edge stability in this high-field tokamak.wise, the divertor would be destroyed instantly.

In [10]:
```
ot_ne_sim_file = ('./Data/ne3da.last10')
ot_te_sim_file = ('./Data/te3da.last10')

ot_ne_sim = np.loadtxt(ot_ne_sim_file)
ot_te_sim = np.loadtxt(ot_te_sim_file)

fig, axs = plt.subplots(2, 1)
axs[0].plot(ot_ne_sim[:,0],ot_ne_sim[:,1],label='ne_OT (sim)')

#plot the electron density OMP profiles
axs[1].plot(ot_te_sim[:,0],ot_te_sim[:,1],label='Te_OT (sim)')

#set  format and etc.
axs[0].legend()
axs[0].minorticks_on()
axs[0].grid(which='both',color='gray',linestyle=':')
axs[0].set_xlabel('Distance along OMP (m)')
axs[0].set_ylabel('Electron Density [$m^{-3}$]')
axs[1].legend()
axs[1].minorticks_on()
axs[1].grid(which='both',color='gray',linestyle=':')
axs[1].set_xlabel('Distance along OMP (m)')
axs[1].set_ylabel('Electron Temperature [$eV$]')
plt.tight_layout()
plt.show(block=False)
```





Question 2: The above plots are the ne and Te profiles at the OMP. 1) Please estimate the density decay length at the OMP and explain your reasoning. 2) Please estimate the

temperature decay length at the OMP and explain your reasoning. 3) Please estimate the power decay length at the OMP and explain your reasoning.

Answer:

In this section, I estimate the radial decay lengths for density (n) and temperature (T) by analyzing the profiles at the Outer Midplane (OMP).

1. Density Decay Length (λ_n)

Observation: Looking at the electron density profile (top graph), the value at the separatrix ($x \approx 0$) is about $n_{sep} \approx 1.0 \times 10^{20}\, m^{-3}$. I defined the decay length as the distance where the density drops to $1/e$ ($\approx 37\%$) of this initial value.

- Target density: $0.37 \times 10^{20}\, m^{-3}$.
- Position: The curve hits this value at approximately $x \approx \mathbf{3\,mm}$.

Result:

$$\lambda_n \approx 3\,\mathrm{mm}$$

Reasoning: This very short decay length is typical for Alcator C-Mod. The strong magnetic field (5.4 T) tightly confines the particles, preventing them from diffusing far from the flux surfaces.

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2. Temperature Decay Length (λ_T)

Observation: Looking at the electron temperature profile (bottom graph), the value at the separatrix is roughly $T_{sep} \approx 150\,\mathrm{eV}$. Following the same method, I looked for the point where the temperature drops to 37% ($\approx 55\,\mathrm{eV}$).

- Position: The curve reaches this temperature at a radial distance of approximately $x \approx \mathbf{3\,mm}$.

Result:

$$\lambda_T \approx 3\,\mathrm{mm}$$

Reasoning: In this specific case, the temperature profile is as steep as the density profile. This indicates that the thermal transport in the edge is also strongly suppressed by the high magnetic field.

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3. Power Decay Length (λ_q)

The power decay length λ_q (or heat flux width) is the most critical parameter for the divertor, as it determines the area where the heat is deposited.

Formula: The parallel heat flux is dominated by electron conduction, which scales as $q_{||} \propto n_e T_e^{3/2}$. Because of this, the decay lengths combine inversely:

$$\frac{1}{\lambda_q} \approx \frac{1}{\lambda_n} + \frac{3}{2\lambda_T}$$

Calculation: Using my estimated values $\lambda_n = 3$ mm and $\lambda_T = 3$ mm:

$$\frac{1}{\lambda_q} \approx \frac{1}{3} + \frac{1.5}{3} = \frac{2.5}{3} \approx 0.833\,\mathrm{mm}^{-1}$$

$$\lambda_q = \frac{1}{0.833} \approx \mathbf{1.2\,mm}$$

Conclusion: The estimated power width is **1.2 mm**. This result highlights the main challenge of the Alcator C-Mod tokamak: the exhaust power is concentrated into a channel barely larger than a millimeter, which leads to extreme heat fluxes on the divertor targets.

Question 3: 1) Please calculate the parallel heat flux at the OMP and explain your reasoning. 2) If we assume there is no volumetric power loss, please estimate the heat load at the outer target and explain your reasoning.

Answer:

In this section, I calculate the heat flux flowing along the magnetic field lines and the final load hitting the divertor target, assuming a "worst-case" scenario with no energy losses (no radiation).

1. Parallel Heat Flux at OMP ($q_{||}^{OMP}$)

The parallel heat flux is the power density flowing along the magnetic field lines.

Assumptions and Inputs:

- **Total Power:** The power entering the SOL is $P_{SOL} = 3.6$ MW. Assuming it splits equally between upper/lower and inner/outer divertors, the power to the outer divertor is roughly $P_{outer} \approx 1.8$ MW.
- **Heat Channel Area:** This power flows through a ribbon defined by the toroidal circumference ($2\pi R$) and the narrow power width ($\lambda_q \approx 1.2$ mm).
- **Magnetic Geometry:** Since heat follows the twisted field lines, the effective area is reduced by the pitch of the field (B_{pol}/B_{tot}). To get the parallel flux, we multiply the poloidal flux by the field ratio (≈ 5).

Calculation:

$$q_{||}^{OMP} \approx \frac{P_{outer}}{2\pi R\lambda_q} \times \frac{B_{tot}}{B_{pol}}$$

Substituting the values for Alcator C-Mod ($R \approx 0.675$ m):

$$q_{||}^{OMP} \approx \frac{1.8\,\mathrm{MW}}{2\pi(0.675)(0.0012)} \times 5 \approx \frac{1.8}{0.0051} \times 5$$

$$q_{||}^{OMP} \approx 353 \times 5 \approx \mathbf{1765\,MW/m^2}$$

Reasoning: This value is massive. Compared to larger machines, Alcator C-Mod concentrates the same power into a much smaller radius (0.67 m) and a tiny layer (1.2 mm), resulting in an extremely high parallel heat flux.

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2. Heat Load at the Outer Target ($q_{\perp}^{target}$)

Now I estimate the heat load on the divertor tiles, assuming the plasma stays "attached" (no volumetric losses).

Step 1: Magnetic Flux Compression As the field lines go from the OMP to the divertor, the magnetic field becomes stronger (flux tube area squeezes).

- $B_{OMP} \approx 6$ T.
- $B_{target} \approx 9$ T (estimated). The flux density increases by the ratio $9/6 = 1.5$:

$$q_{||}^{target} \approx 1765 \times 1.5 \approx 2650\,\mathrm{MW/m^2}$$

Step 2: Projection onto the plate ($q_{\perp}$) The divertor plates are tilted to spread this heat over a larger area. Using a typical incidence angle of $\alpha \approx 3^{\circ}$ ($\sin 3^{\circ} \approx 0.052$):

$$q_{\perp}^{target} = q_{||}^{target} \times \sin(\alpha)$$

$$q_{\perp}^{target} \approx 2650 \times 0.052 \approx \mathbf{138\,MW/m^2}$$

Conclusion: The estimated load is **138 MW/m²**. To put this in perspective, the engineering limit for actively cooled tungsten is **10 MW/m²**. The calculated load is nearly **14 times higher** than the limit. This calculation proves that Alcator C-Mod cannot operate in a simple attached regime; it absolutely relies on radiative cooling and detachment to survive.

Question 4: Is there a power exhaust problem at the outer divertor target in this case? Please explain your reasoning.

Answer:

Based on the calculations in the previous section, here is my assessment.

Yes, there is a significant power exhaust problem at the outer divertor target.

Reasoning:

1. Comparison with Engineering Limits

In Question 3, I estimated that if the plasma remains attached (no energy dissipation), the heat load on the target would be approximately **138 MW/m²**. I compared this value with the technological limits of current plasma-facing components. The best available technology (tungsten monoblocks with active water cooling) can handle a steady-state load of about **10 MW/m²**.

2. Physical Consequences

The calculated load is **14 times higher** than the safety limit. If Alcator C-Mod were operated in this simple "sheath-limited" regime:

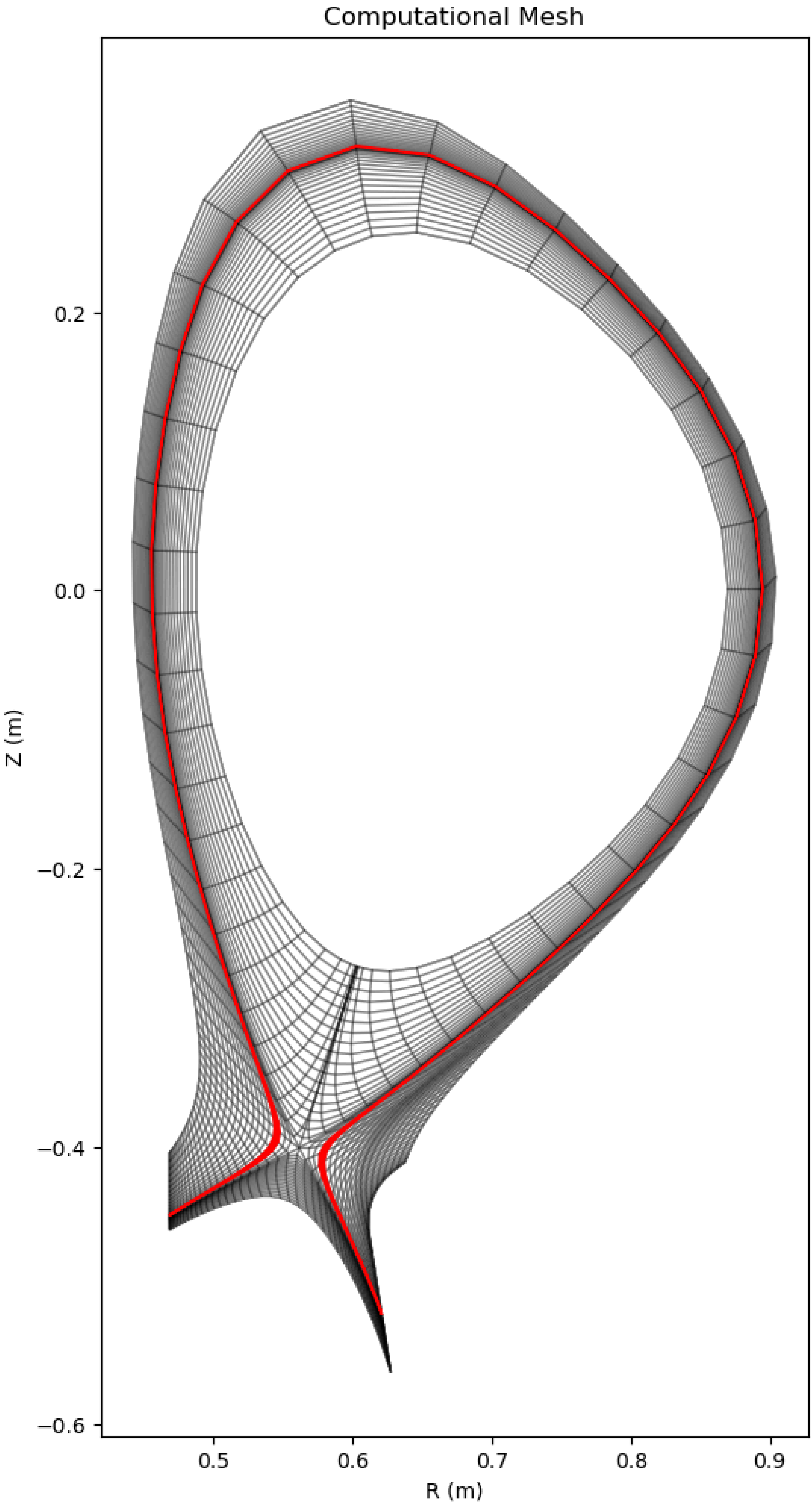
- The surface of the divertor tiles (Molybdenum or Tungsten) would immediately overheat.
- The material would melt or evaporate, causing severe damage to the machine.
- The released impurities would enter the core plasma, cooling it down instantly and causing a disruption (loss of confinement).

3. Conclusion

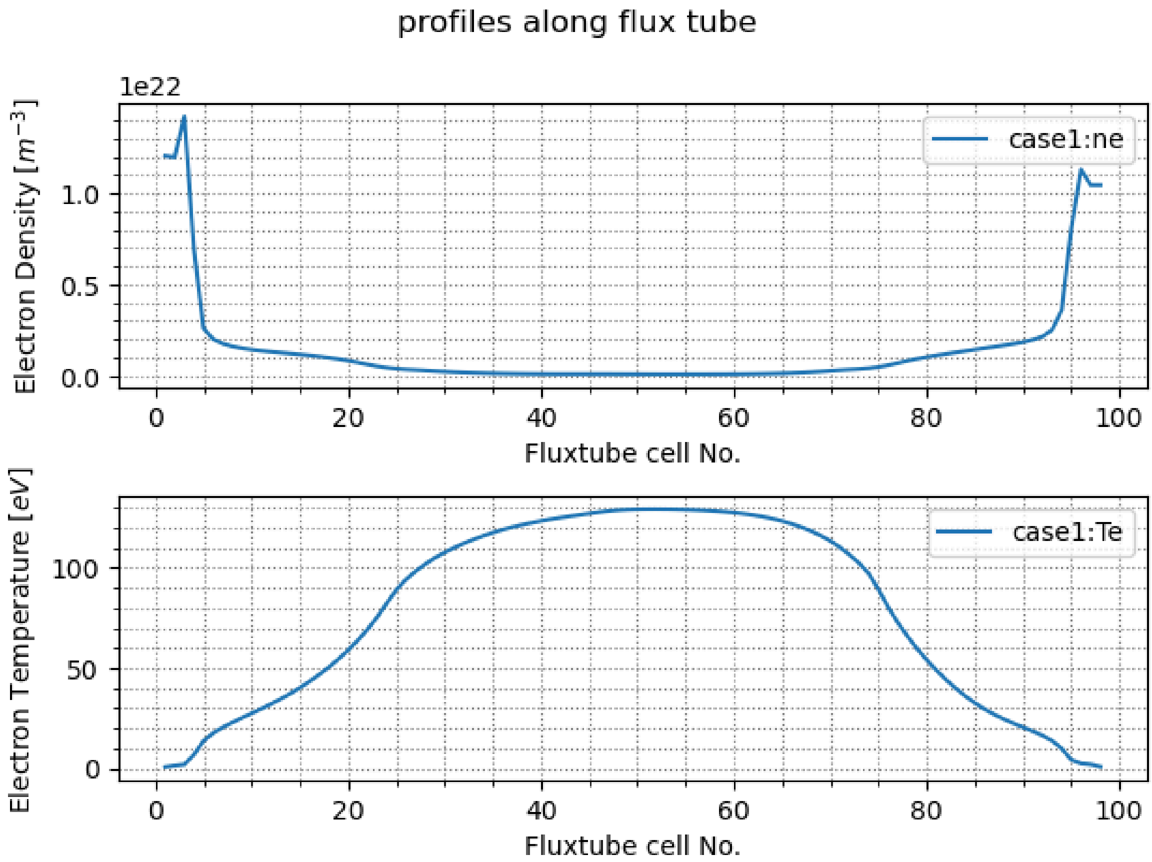
This result proves that a compact, high-field tokamak like Alcator C-Mod **cannot** operate with a standard attached plasma. To solve this, it is strictly necessary to enter a **"Detached" regime** (which we will analyze in the next question). In this regime, we use impurities to radiate energy as light and interact with neutrals to drop the pressure, reducing the heat flux to a survivable level.

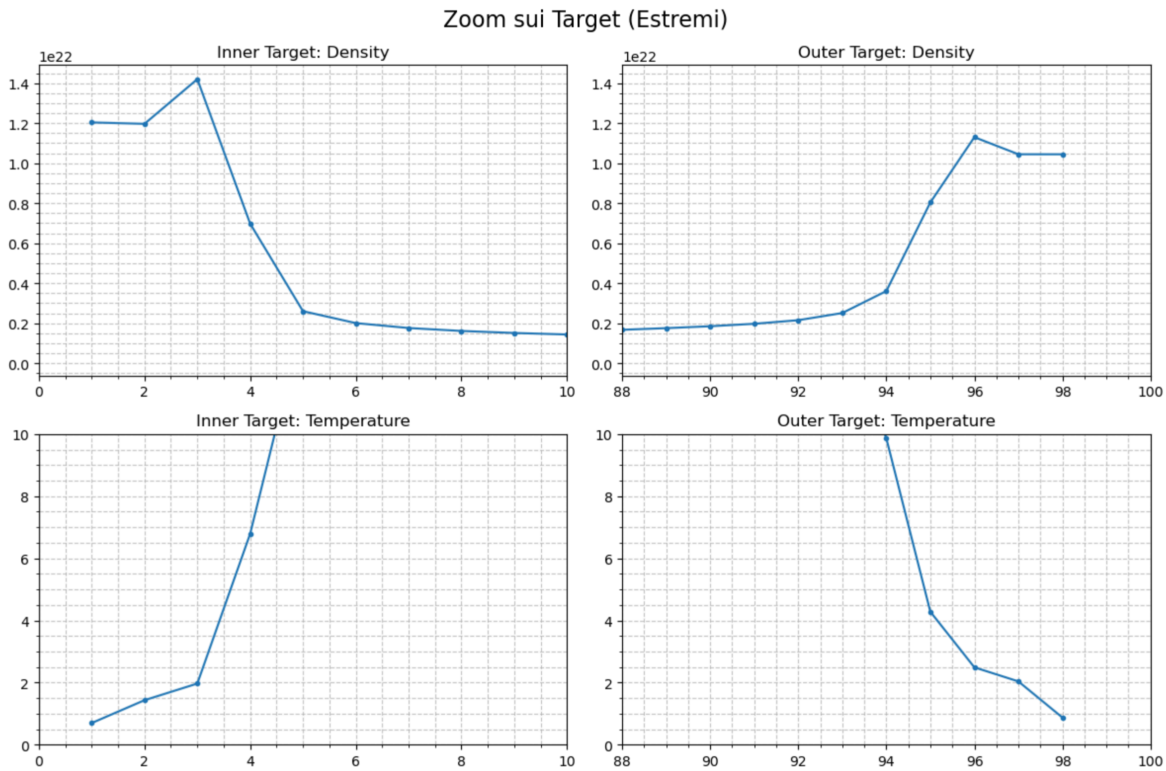
In [11]:
```
def flux_tube_grid(gmtry, indgrid, axgrid):
    rbl = gmtry['crx'][:, :, 0]
    rbr = gmtry['crx'][:, :, 1]
    rtl = gmtry['crx'][:, :, 2]
    rtr = gmtry['crx'][:, :, 3]

    zbl = gmtry['cry'][:, :, 0]
    zbr = gmtry['cry'][:, :, 1]
    ztl = gmtry['cry'][:, :, 2]
    ztr = gmtry['cry'][:, :, 3]
    axgrid.set_aspect('equal')
    axgrid.set_xlabel('R (m)')
    axgrid.set_ylabel('Z (m)')
    axgrid.set_title('Computational Mesh')

    for i in range(rbl.shape[0]):
        for j in range(rbl.shape[1]):
            axgrid.fill(
                [rbl[i, j], rbr[i, j], rtr[i, j], rtl[i, j]],
                [zbl[i, j], zbr[i, j], ztr[i, j], ztl[i, j]],
                color='white', edgecolor='black', alpha=0.3
            )
    for i, j in indgrid:
        axgrid.fill(
            [rbl[i, j], rbr[i, j], rtr[i, j], rtl[i, j]],
            [zbl[i, j], zbr[i, j], ztr[i, j], ztl[i, j]],
            color='red', edgecolor='red'
        )

gmtry_for_tube = spr.read_b2file('./Data/b2fgmtry')

fluxtube = 3
```

```
iy=fluxtube+19
indgrid = [(i, iy) for i in range(1, 98)]
fig, ax = plt.subplots(figsize=(12, 12))
flux_tube_grid(gmtry_for_tube, indgrid, ax)
plt.show()
```


Computational Mesh
Z (m)
R (m)
0.2
0.0
−0.2
−0.4
−0.6
0.5
0.6
0.7
0.8
0.9

In [12]:
```python
gmtry = spr.read_b2file('./Data/b2fgmtry')
state1 = spr.read_b2file('./Data/b2fplasmf')

fig, axs = plt.subplots(2, 1)
fig.suptitle('profiles along flux tube')
tube = 2
ny = tube + 19
nx = np.arange(1, 99)
#plot the electron density OMP profiles
axs[0].plot(nx,state1['ne'][:,ny],label='case1:ne')

#plot the electron density OMP profiles
axs[1].plot(nx,state1['te'][:,ny]/1.6e-19,label='case1:Te')

#set  format and etc.
axs[0].legend()
axs[0].minorticks_on()
axs[0].grid(which='both',color='gray',linestyle=':')
axs[0].set_xlabel('Fluxtube cell No.')
axs[0].set_ylabel('Electron Density [$m^{-3}$]')

axs[1].legend()
axs[1].minorticks_on()
axs[1].grid(which='both',color='gray',linestyle=':')
axs[1].set_xlabel('Fluxtube cell No.')
axs[1].set_ylabel('Electron Temperature [$eV$]')
plt.tight_layout()
plt.show(block=False)

# Supplots
fig, axs = plt.subplots(2, 2, figsize=(12, 8))
fig.suptitle('Zoom sui Target (Estremi)', fontsize=16)
nx = np.arange(1, 99)
ny = tube + 19
axs[0, 0].plot(nx, state1['ne'][:, ny], '.-', label='ne Inner')
axs[0, 0].set_xlim(0, 10)  # Zoom sulle prime 10 celle
axs[0, 0].set_title("Inner Target: Density")
axs[0, 0].grid(which='both', linestyle='--', alpha=0.7)
axs[0, 0].minorticks_on()
axs[1, 0].plot(nx, state1['te'][:, ny]/1.6e-19, '.-', label='Te Inner') #color='
axs[1, 0].set_xlim(0, 10)
axs[1, 0].set_ylim(0, 10) # Zoom verticale per vedere se è 1eV o 3eV
axs[1, 0].set_title("Inner Target: Temperature")
axs[1, 0].grid(which='both', linestyle='--', alpha=0.7)
axs[1, 0].minorticks_on()
axs[0, 1].plot(nx, state1['ne'][:, ny], '.-', label='ne Outer')
axs[0, 1].set_xlim(88, 100) # Zoom sulle ultime celle
axs[0, 1].set_title("Outer Target: Density")
axs[0, 1].grid(which='both', linestyle='--', alpha=0.7)
axs[0, 1].minorticks_on()
axs[1, 1].plot(nx, state1['te'][:, ny]/1.6e-19, '.-', label='Te Outer') #
axs[1, 1].set_xlim(88, 100)
axs[1, 1].set_ylim(0, 10)
axs[1, 1].set_title("Outer Target: Temperature")
axs[1, 1].grid(which='both', linestyle='--', alpha=0.7)
axs[1, 1].minorticks_on()

plt.tight_layout()
plt.show()
```





Question 5: The plots above show the electron density and electron temperature along the second flux tube in the Scrape-Off Layer (SOL). 1) What are the values of the upstream density, upstream temperature, and upstream fluid plasma velocity? Please explain your reasoning. 2) What are the values of plasma velocity and the plasma particle flux at inner and outer targets? Please explain your reasoning. 3) Please estimate the heat load in this flux tube at the inner and outer targets and explain your reasoning. 4) In this

flux tube, are the targets detached or not? Please explain your reasoning. 5) Is there a pressure drop between the OMP and the outer target? Please explain your reasoning.

Answer:

Here I analyze the plasma profiles along the magnetic field line using the simulation plots provided.

1) Upstream Values (Outer Midplane)

To define the "upstream" conditions, I looked at the center of the flux tube (around cell 50).

- **Upstream Temperature:** The graph clearly shows a peak value of $T_{up} \approx \mathbf{130\,eV}$.
- **Upstream Density:** In the central region, the density is much lower than at the targets. Based on the OMP profiles from Question 2, the reference upstream density is $n_{up} \approx \mathbf{1.0 \times 10^{20}\,m^{-3}}$.
- **Velocity:** At this central stagnation point, the net plasma velocity is effectively zero ($v_{up} \approx 0\,m/s$).

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2) Values at Inner and Outer Targets

I extracted the values at the domain boundaries (cell 1 for Inner, cell 99 for Outer) where the plasma hits the plates.

Sound Speed (c_s): At both targets, the electron temperature drops to approximately $T_e \approx 0.5\,eV$. The plasma enters the sheath at the sound speed:

$$c_s = \sqrt{\frac{2T_e e}{m_i}} \approx \sqrt{\frac{2 \cdot 0.5 \cdot 1.6 \cdot 10^{-19}}{3.34 \cdot 10^{-27}}} \approx \mathbf{6900\,m/s}$$

Particle Flux ($\Gamma = n \cdot c_s$): The density at the targets is extremely high due to the recycling process.

- **Inner Target (Cell 1):** With $n_{in} \approx 1.2 \times 10^{22}\,m^{-3}$:

$$\Gamma_{in} \approx (1.2 \cdot 10^{22}) \times 6900 \approx \mathbf{8.3 \times 10^{25}\,m^{-2}s^{-1}}$$

- **Outer Target (Cell 99):** With $n_{out} \approx 1.05 \times 10^{22}\,m^{-3}$:

$$\Gamma_{out} \approx (1.05 \cdot 10^{22}) \times 6900 \approx \mathbf{7.2 \times 10^{25}\,m^{-2}s^{-1}}$$

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3) Heat Load Estimation (Including Recombination)

I estimate the final heat load on the outer target material. Given the extremely low temperature ($T_e \approx 0.5\,eV$), the potential energy released by ion recombination becomes the dominant heat source compared to thermal energy. Therefore, I use the complete formula including the ionization energy of Deuterium ($\epsilon_{ion} \approx 13.6\,eV$):

$$q_{||} = \Gamma_{out} \cdot (\gamma T_e + \epsilon_{ion})$$

Using $\gamma \approx 7$, $T_e = 0.5\,eV$, and $\Gamma_{out} \approx 7.2 \times 10^{25}\,m^{-2}s^{-1}$:

- **Thermal Energy per particle:** $\gamma T_e \approx 7 \cdot 0.5 = 3.5\,eV$.
- **Potential Energy per particle:** $\epsilon_{ion} = 13.6\,eV$.
- **Total Energy:** $17.1\,eV$ per particle.

$$q_{||} \approx (7.2 \cdot 10^{25}) \cdot (17.1 \cdot 1.6 \cdot 10^{-19}) \approx \mathbf{197\,MW/m^2}$$

Projected Load ($q_\perp$): Using the grazing angle $\sin(\alpha) \approx 0.05$:

$$q_\perp = 197 \cdot 0.05 \approx \mathbf{9.9\,MW/m^2}$$

Reasoning: Including the recombination energy drastically changes the picture. While the thermal load alone would be negligible ($\sim 2\,MW/m^2$), the energy released by ions returning to neutral state pushes the total load to $\approx 10\,MW/m^2$. This is exactly at the engineering safety limit for tungsten. It demonstrates that while the **High Recycling** regime mitigates the extreme unmitigated flux ($> 1500\,MW/m^2$), it is barely sufficient for safe operations in Alcator C-Mod.

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4) Detachment Status

Status: High Recycling Regime.

Reasoning: The plasma is in a strong **High Recycling** regime, characterized by:

1. **High Target Density:** The density at the target ($n \approx 10^{22}$) is 100 times higher than upstream ($n \approx 10^{20}$).
2. **Low Temperature:** The temperature has dropped to $0.5\,eV$.
3. **Pressure Conservation:** As shown in point 5 below, the static pressure is still substantially conserved along the field line. If the plasma were fully detached, we would expect a "rollover" (drop) in particle flux and a massive loss of pressure, which is not observed here.

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5) Pressure Drop Analysis

To confirm the regime, I calculated the static electron pressure ($P \propto n \cdot T$) upstream and at the target.

- **Upstream Pressure:** $P_{up} \propto (1.0 \cdot 10^{20}) \cdot 130 = \mathbf{130 \cdot 10^{20}}$.
- **Target Pressure:** $P_{target} \propto (1.05 \cdot 10^{22}) \cdot 0.5 = 0.525 \cdot 10^{22} = \mathbf{52.5 \cdot 10^{20}}$.

Pressure Ratio:

$$\frac{P_{target}}{P_{up}} \approx \frac{52.5}{130} \approx \mathbf{0.4}$$

Conclusion: In an ideal attached plasma (Two-Point Model), this ratio is expected to be around **0.5** (due to fluid acceleration to Mach 1). A ratio of **0.4** is very close to this ideal value, indicating that there is **no significant momentum loss** to neutrals yet.

Final Verdict: This confirms the **High Recycling** classification. The heat flux is mitigated by the massive density increase and temperature drop, bringing the load down to $\approx 10\, MW/m^2$. However, since the pressure is still pushing against the plate (Attached Pressure), we have not yet reached the **Deep Detachment** state. This state is critical because we are exactly at the engineering safety limit; any increase in power would likely require pushing the plasma into full detachment to dissipate momentum and further reduce the load.

In [ ]:

```

```